

Customer Insights (Behavior, Segmentation, Loyalty)

1. Which customer type generates the highest average revenue per booking?

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```
select CustomerType, avg(Revenue) as avg_revenue
from HotelBookings_replica
GROUP BY CustomerType
ORDER BY avg_revenue DESC;
```

CustomerType	avg_revenue
Contract	381.547872
Transient	253.275907
Group	213.787001
Transient-Party	207.978002

2. How does the average number of nights differ across customer types (e.g., Transient vs. Group)?

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```
select CustomerType, AVG(Nights) as avg_nights
from HotelBookings_replica
group by CustomerType
order by avg_nights asc;
```

CustomerType	avg_nights
Group	2
Transient	3
Transient-Party	3
Contract	5

3. Which countries have the highest average daily rate (ADR)?

/*Which countries have the highest average daily rate (ADR)?*/

```

select top(10) Country, avg(AvgDailyRate) as avg_of_avg_daily_Rate
from HotelBookings_replica
group by Country
order by avg_of_avg_daily_Rate desc;

```

Country	avg_of_avg_daily_Rate
Djibouti	273.000000
Anguilla	265.000000
Andorra	202.652857
Mayotte	177.750000
New Caledonia	175.500000
Georgia	169.015909
Comoros	165.305000
Faroe Islands	155.200000
Philippines	154.355750
Gibraltar	150.402222

4. Identify the top 10 countries with the highest booking volume. How does their average cancellation rate compare?

```

CREATE NONCLUSTERED INDEX IX_Country_Cancelled
ON HotelBookings_replica (Country, Cancelled);

```

```

declare @total int
select @total= count(Cancelled) from HotelBookings_replica where
Cancelled=1;

```

```

with tab1 as(
select Country,(CAST(COUNT(Cancelled) AS decimal) / @total)*100 as
booking_percentages
from HotelBookings_replica
where Cancelled=1
group by Country
),

```

```

tab2 as (
select Country, count(BookingID) as Booking_count
from HotelBookings_replica
group by Country
)

```

```

select top(10)
tab2.Country,tab2.Booking_count,isnull(tab1.booking_percentages,0) as
cancellation_Rate
from tab2
left join tab1 on tab1.Country=tab2.Country
order by tab2.Booking_count desc;

```

Country	Booking_count	cancellation_Rate
Portugal	48590	62.22639290800
United Kingdom	12129	5.54676193900
France	10415	4.37319102700
Spain	8568	4.92266642500
Germany	7287	2.75416063600
Italy	3766	3.01420043400
Ireland	3375	1.88133140300
Belgium	2342	1.07181620800
China	2280	1.61903039000
Brazil	2224	1.87680897200

5. Which customer type has the highest cancellation rate and average lead time?

```

declare @total int
select @total= count(Cancelled) from HotelBookings_replica where
Cancelled=1;

select CustomerType,(CAST(COUNT(Cancelled) AS decimal) / @total)*100 as
cancellation_percentages
from HotelBookings_replica
where Cancelled=1
group by CustomerType
order by cancellation_percentages desc
;

```

CustomerType	cancellation_percentages
Transient	82.56602749600
Transient-Party	14.44690665700
Contract	2.85365412400
Group	0.13341172200

```

select CustomerType,avg(LeadTime) as avg_lead_time
from HotelBookings_replica
where Cancelled=1
group by CustomerType
order by avg_lead_time desc
;

```

CustomerType	avg_lead_time
Contract	221
Transient-Party	165
Group	144
Transient	138

6. What is the median lead time for each customer type?

```

select distinct CustomerType,
PERCENTILE_CONT(0.5) within group (order by LeadTime asc)
over(partition by CustomerType) as Median
from HotelBookings_replica;

```

CustomerType	Median
Group	12
Transient	61
Transient-Party	102
Contract	114

7. Cluster countries by similar booking behaviors: number of guests, lead time, and ADR (k-means possible after SQL pre-aggregation).
8. Calculate the percentage of bookings with more than 3 guests and analyze their revenue impact.

```

create nonclustered index ix_guest
on HotelBookings_replica(Guests);

```

```
select
(count(case when Guests>3 then BookingID else null end)/cast(count(*) as
decimal))*100 as Booking_percentages,
sum(case when Guests>3 then Revenue else 0 end) as total_Rev,
sum(case when Guests>3 then RevenueLoss else 0 end) as total_Rev_loss
from HotelBookings_replica;
```

Booking_percentages	total_Rev	total_Rev_loss
3.4198844124298517400	1548398.24	-1480744.52

Although large group bookings are rare, they involve high revenue and high risk. Businesses should analyze cancellation reasons and consider offering better support or terms to reduce revenue loss.

Revenue Management (Profitability, ADR, Revenue Loss)

1. Calculate total revenue, revenue loss, and net revenue monthly.

```
WITH tab1 AS (
  SELECT MONTH(ArrivalDate) AS Month_number, SUM(Revenue) AS
Total_Rev
  FROM HotelBookings_replica
  GROUP BY MONTH(ArrivalDate)
),
tab2 AS (
  SELECT MONTH(StatusUpdate) AS Month_number, SUM(RevenueLoss) AS
Total_RevLoss
  FROM HotelBookings_replica
  GROUP BY MONTH(StatusUpdate)
)
SELECT
  tab1.Month_number,
  tab1.Total_Rev,
  tab2.Total_RevLoss,
  (tab1.Total_Rev + ISNULL(tab2.Total_RevLoss, 0)) AS Net_Rev
FROM tab1
LEFT JOIN tab2
  ON tab1.Month_number = tab2.Month_number
ORDER BY tab1.Month_number ASC;
```

Month_number	Total_Rev	Total_RevLoss	Net_Rev
1	860798.84	-1297461.78	-436662.94
2	1292835.48	-1286831.32	6004.16
3	1913763.65	-1467601.99	446161.66
4	2567842.79	-1346559.80	1221282.99
5	2883721.64	-1389140.46	1494581.18
6	3195043.00	-1271984.42	1923058.58
7	4251957.07	-1413202.79	2838754.28
8	5052890.46	-942814.55	4110075.91

January saw a net loss due to high cancellations, but revenue steadily improved from February, peaking between May and August with August showing the highest net revenue. The last quarter remained profitable but at a lower level. To improve performance, focus marketing efforts on the summer months, implement stricter or smarter cancellation policies in early months, and apply dynamic pricing to boost low-demand periods.

*/*This is a classic example for creating temporary table*/*

-- tab3 as temp table

```
SELECT MONTH(ArrivalDate) AS Month_number, SUM(Revenue) AS Total_Rev
INTO #tab3
FROM HotelBookings_replica
GROUP BY MONTH(ArrivalDate);
```

-- tab4 as temp table

```
SELECT MONTH(StatusUpdate) AS Month_number, SUM(RevenueLoss) AS
Total_RevLoss
INTO #tab4
FROM HotelBookings_replica
GROUP BY MONTH(StatusUpdate);
```

-- Optional: Add index on temp tables

```
CREATE NONCLUSTERED INDEX ix_month_tab1 ON #tab3(Month_number);
CREATE NONCLUSTERED INDEX ix_month_tab2 ON #tab4(Month_number);
```

-- Final join query

```
SELECT
    t1.Month_number,
    t1.Total_Rev,
    t2.Total_RevLoss
FROM #tab3 t1
LEFT JOIN #tab4 t2 ON t1.Month_number = t2.Month_number
ORDER BY t1.Month_number;
```

```
select * from #tab1 t
order by t.Month_number;
```

2. What is the average revenue loss due to cancellations per month?

```
select month(StatusUpdate) as Month_Number,
avg(RevenueLoss) as AvgRevLoss
from HotelBookings_replica
where Cancelled = 1
group by month(StatusUpdate)
order by month(StatusUpdate) asc;
```

Month_Number	AvgRevLoss
1	-214.067279
2	-295.280247
3	-384.189002
4	-388.057579
5	-442.683384
6	-436.807836
7	-323.461384
8	-350.749460
9	-287.888352
10	-162.134248

The analysis shows that the average revenue loss per cancelled booking is highest between April and June, peaking in May (-442.68), while the lowest losses occur in October (-162.13). This suggests cancellations in mid-year months are more financially damaging—likely due to higher-value bookings in peak seasons. To reduce impact, the business should consider stricter cancellation policies or upfront deposits during high-loss months, and analyze booking sources or customer types contributing to these losses.

3. Which distribution channel is associated with the highest revenue loss rate?

```
select DistributionChannel,abs(SUM(RevenueLoss)) as Total_revLoss
from HotelBookings_replica
group by DistributionChannel
```

order by Total_revLoss desc;

DistributionChannel	Total_revLoss
Online Travel Agent	10782870.82
Offline Travel Agent	1110542.27
Direct	1033280.81
Corporate	195598.69
Undefined	607.50

The Online Travel Agent channel incurs the highest revenue loss by a large margin, indicating that cancellations or issues through this channel significantly impact overall profitability. To reduce losses, the business should strengthen cancellation policies, improve booking verification, and negotiate better terms with online agents. Additionally, monitoring booking patterns and customer behavior in this channel can help identify risk factors early and implement targeted interventions.

4. Identify the top 10 days with the highest net revenue.

```
select top(10) ArrivalDate,SUM(Net_Rev) as actual_netRev
from(
select ArrivalDate,(Revenue+RevenueLoss) as Net_Rev
from HotelBookings_replica
) tab1
group by ArrivalDate
order by actual_netRev desc;
```


ArrivalDate	actual_netRev
2015-09-25	159298.24
2015-09-24	107592.08
2016-11-07	106252.31
2017-05-15	92525.07
2017-06-08	81858.77
2016-06-12	80383.56
2015-10-05	76375.73
2015-09-26	68999.04
2016-06-15	64890.00
2016-09-26	64625.27

The top 10 days show significant net revenue peaks, indicating these dates likely had high booking volumes or premium pricing. To capitalize on this, analyze factors like seasonality, special events, or promotions driving these spikes and aim to replicate successful strategies on similar future dates for sustained revenue growth.

5. Which country brings in the highest revenue per night on average?

```
select Country, (SUM(Revenue)/SUM(Nights)) as RevPerNights
from HotelBookings_replica
where Nights>0
group by Country
order by RevPerNights desc;
```

Country	RevPerNights
Djibouti	273.000000
Anguilla	265.000000
New Caledonia	175.500000
Comoros	165.305000
El Salvador	152.566666
Myanmar	145.000000
French Polynesia	144.000000
Palau	136.290000
Antarctica	134.250000
Aruba	132.531428

Countries with the highest average revenue per night likely host premium or long-staying guests. Focus marketing, upselling, and loyalty efforts on these countries to boost profitability and retain high-value customers.

- Calculate the coefficient of variation (standard deviation ÷ mean) of ADR across months to assess rate volatility.

```
select month(ArrivalDate) as MonthNumber,
(STDEVP(AvgDailyRate)*1.0/AVG(AvgDailyRate)) as CoefficientVariance
from HotelBookings_replica
where AvgDailyRate>0
group by month(ArrivalDate)
order by month(ArrivalDate) asc;
```

MonthNumber	CoefficientVariance
1	0.370786337860599
2	0.382052721452602
3	0.765800060060162
4	0.361550287843298
5	0.371404985479757
6	0.3134513879639
7	0.391052820468894
8	0.423508769682588
9	0.387456553905641
10	0.382219355196064
11	0.420297891219196
12	0.486283696414012

The monthly price variation (CV) is mostly steady, but March shows much more ups and downs in prices. Check March to understand why prices change a lot and adjust pricing if needed. Other months look stable.

7. Which customer type shows the greatest variability in ADR?

```
select CustomerType,STDEVP(AvgDailyRate) as StandardDeviations
from HotelBookings_replica
group by CustomerType
order by StandardDeviations desc;
```

CustomerType	StandardDeviations
Transient	53.340994121719
Group	50.2204702136784
Transient-Party	36.8892354120959
Contract	33.6834715135587

Transient customers have the highest ADR variability, showing diverse spending patterns. Focus on analyzing their behavior to optimize pricing and boost revenue.

8. Compute revenue per guest per booking and analyze trends across seasons.

```
with tab1 as (  
  select MONTH(ArrivalDate) as Month_Number, (SUM(Revenue)/SUM(Guests)) as  
  RevPerGuest  
  from HotelBookings_replica  
  where Guests>0  
  group by MONTH(ArrivalDate)  
)  
  
tab2 as (  
  select Month_Number,  
  case  
    WHEN Month_Number BETWEEN 3 AND 5 THEN 'Spring'  
    WHEN Month_Number BETWEEN 6 AND 8 THEN 'Summer'  
    WHEN Month_Number BETWEEN 9 AND 11 THEN 'Autumn'  
    else 'Winter'  
  end as Seasons  
  from tab1  
)  
select tab2.Month_Number,tab2.Seasons,tab1.RevPerGuest  
from tab2  
left join tab1 on tab2.Month_Number=tab1.Month_Number  
order by tab2.Month_Number;
```

Month_Number	Seasons	RevPerGuest
1	Winter	81.757048
2	Winter	84.447395
3	Spring	103.807910
4	Spring	116.682910
5	Spring	128.380441
6	Summer	148.703067
7	Summer	153.986551
8	Summer	164.155206
9	Autumn	141.551492
10	Autumn	107.162332
11	Autumn	102.591702
12	Winter	93.146598

Revenue per guest is lowest in Winter (around \$82–93) and steadily rises through Spring, peaking in Summer (up to \$164), then slightly declines in Autumn. This suggests Summer is the most profitable season per guest, likely

due to peak travel. To maximize profits, focus marketing, premium packages, and upselling efforts in Summer, while using discounts or promotions in Winter to boost occupancy and guest spending.

9. Compare ADR before and after cancellations by customer type.

```
with cte as (  
  select CustomerType,avg(AvgDailyRate) as Total_AVD_before_cancell  
  from HotelBookings_replica  
  where Cancelled=0  
  GROUP BY CustomerType),  
cte2 as (  
  select CustomerType,avg(AvgDailyRate) as Total_AVD_after_cancell  
  from HotelBookings_replica  
  where Cancelled=1  
  GROUP BY CustomerType)  
  
select c2.CustomerType,  
c.Total_AVD_before_cancell,c2.Total_AVD_after_cancell  
from cte2 c2  
full outer join cte c on c.CustomerType=c2.CustomerType;
```

CustomerType	Total_AVD_before_cancell	Total_AVD_after_cancell
Transient	104.776032	110.267542
Transient-Party	88.180894	79.936097
Group	82.450057	92.606440
Contract	91.469150	78.809928

Transient and Group customers cancel more high-ADR bookings, causing greater revenue risk, while Contract and Transient-Party cancel lower-value stays. Focus on retaining Transient and Group customers through flexible policies or loyalty perks to reduce premium booking losses.



Operational Efficiency (Capacity, Flow, Optimization)

1. Which day of the week has the highest average arrival count?

```
with cte as(
select DATENAME(WEEKDAY, ArrivalDate) AS Day_of_week,
ArrivalDate,
COUNT(*) AS Arrival_count
FROM HotelBookings_replica
where Cancelled = 0
group by DATENAME(WEEKDAY, ArrivalDate),ArrivalDate)
```

```
select Day_of_week,AVG(Arrival_count) as Avg_count
from cte
group by Day_of_week
order by Avg_count desc;
```

Day_of_week	Avg_count
Monday	105
Friday	103
Thursday	99
Saturday	96
Wednesday	90
Sunday	84
Tuesday	83

Monday has the highest average arrivals, indicating peak customer activity at the start of the week, while Tuesday has the lowest. Focus marketing and resource allocation on Mondays and Fridays to maximize revenue and optimize staffing, and consider promotions or incentives to boost arrivals on slower days like Tuesday and Sunday.

2. What is the average length of stay per customer type?

```
select CustomerType,avg(Nights) as avg_Stay
from HotelBookings_replica
group by CustomerType;
```

CustomerType	avg_Stay
Group	2
Contract	5
Transient	3
Transient-Party	3

The data shows that Contract customers have the longest average stay (5 nights), while Group, Transient, and Transient-Party customers typically stay for only 2–3 nights. This suggests that Contract customers contribute more stable, long-term revenue, making them ideal for targeted retention strategies. To increase revenue, hotels should focus on extending the stay duration of short-stay segments through personalized offers, bundled packages, or discounts for extended stays.

- Identify overbooked days (bookings > available capacity — assuming a capacity constraint).

-- lets assume, In this hotel it has maximum capacity of 350 bookings.

select ArrivalDate, TotalBooking,

case

when TotalBooking > 350 then 'Overbooked'

else 'RegularBooking'

end as 'BookingCapacity'

from (

select ArrivalDate, COUNT(distinct BookingID) as TotalBooking

from HotelBookings_replica

group by ArrivalDate

) hotel;

ArrivalDate	TotalBooking	BookingCapacity
2016-01-12	69	RegularBooking
2016-12-09	116	RegularBooking
2015-12-20	38	RegularBooking
2016-11-16	131	RegularBooking
2016-02-27	211	RegularBooking
2017-01-24	87	RegularBooking
2016-02-04	105	RegularBooking
2017-01-01	128	RegularBooking
2015-07-27	170	RegularBooking

4. Find the longest lead times for each distribution channel?

```
select DistributionChannel,  
MAX(LeadTime) as LongestLeadTime  
from HotelBookings_replica  
group by DistributionChannel  
order by LongestLeadTime desc;
```

DistributionChannel	LongestLeadTime
Direct	737
Online Travel Agent	629
Offline Travel Agent	532
Corporate	390
Undefined	103

5. Determine the busiest months by total guest count.

```
select top(3) MONTH(ArrivalDate) as MonthNumber,  
sum(Guests) as GuestsNumber  
from HotelBookings_replica  
group by MONTH(ArrivalDate)  
order by GuestsNumber desc  
;
```

MonthNumber	GuestsNumber
8	30771
7	27607
5	22456

6. What is the percentage of bookings with 1-night stays vs. long stays (7+ nights)?

```
DECLARE @ShortStays DECIMAL(10,2);  
DECLARE @LongStays DECIMAL(10,2);  
DECLARE @TotalStays DECIMAL(10,2);
```

```
SELECT @ShortStays = COUNT(*)  
FROM HotelBookings_replica  
WHERE Nights>0 and Nights<7 ;
```

```
SELECT @LongStays = COUNT(*)
```



```
FROM HotelBookings_replica  
WHERE Nights >= 7;
```

```
SELECT @TotalStays = COUNT(*)  
FROM HotelBookings_replica  
WHERE Nights IS NOT NULL;
```

```
SELECT  
    CAST((@ShortStays * 100.0) / @TotalStays AS DECIMAL(10,2)) AS  
ShortStays_Percentage,  
    CAST((@LongStays * 100.0) / @TotalStays AS DECIMAL(10,2)) AS  
LongStays_Percentage;
```

ShortStays_Percentage	LongStays_Percentage
87.75	11.65

A large majority of bookings (87.75%) are for 1- 6 night stays, while only 11.65% are long stays (7+ nights). This suggests guests mostly prefer short stays, which could impact profitability due to high room turnover. To improve revenue, consider offering discounted multi-night packages, loyalty rewards, or value-added services to encourage longer stays.

7. Analyze booking peaks and identify if lead times are sufficient for operational preparation.

```
select MONTH(ArrivalDate) as MonthNumber,AVG(LeadTime) as AvgLeadTime,  
count(BookingID) as NumOfBookings  
from HotelBookings_replica  
group by MONTH(ArrivalDate)  
order by NumOfBookings desc;
```

MonthNumber	AvgLeadTime	NumOfBookings
8	121	13877
7	136	12661
5	117	11791
10	122	11160
4	94	11089
6	128	10939
9	136	10508
3	70	9794
2	47	8068
11	76	6794
12	73	6780
1	45	5929

The data shows that July, August, and September have the highest number of bookings (12K–13K), indicating clear peak demand months. During these periods, the average lead time is also relatively high (121–136 days), suggesting that most customers plan well in advance. This provides the hotel ample time for operational preparation such as staffing, inventory, and room readiness. However, in January and February, bookings are lower with shorter lead times (45–47 days), which may require more flexible operational planning. It's recommended to prioritize resource allocation during summer peaks and boost promotional efforts in off-peak months to increase occupancy.

8. Calculate standard deviation of guest count per day. Identify high-variance days (operational risk).

```
with GuestCount as(
select ArrivalDate, sum(Guests) as TotalGuests
from HotelBookings_replica
group by ArrivalDate
),
GuestStdv as (
select AVG(TotalGuests) as MeanGuest,
STDEV(TotalGuests) as TotalGueststdv
from GuestCount),
```

```

ZScore as (
select g.ArrivalDate, g.TotalGuests,
round(((g.TotalGuests- s.MeanGuest)/s.TotalGueststdv),3) as Zscore
from GuestCount g
cross join GuestStdv s
)

```

```

select z.ArrivalDate,z.TotalGuests,z.Zscore,
case
    when z.Zscore>2 then 'Unusually high guest'
    when z.Zscore<-2 then 'Unusually low guest'
    else 'Normal'
end as 'GuestQuantity'
from ZScore z
order by z.Zscore desc;

```

ArrivalDate	TotalGuests	Zscore	GuestQuantity
2015-12-05	910	4.785	Unusually high guest
2017-04-29	694	3.102	Unusually high guest
2016-10-28	686	3.04	Unusually high guest
2015-10-16	662	2.853	Unusually high guest
2015-08-14	659	2.829	Unusually high guest
2017-05-25	657	2.814	Unusually high guest
2016-03-24	655	2.798	Unusually high guest
2017-06-08	650	2.759	Unusually high guest
2016-11-07	648	2.743	Unusually high guest
2016-06-17	640	2.681	Unusually high guest
2015-09-18	631	2.611	Unusually high guest

The data shows a clear distinction between "Unusually high guest" days and "Normal" guest days based on significantly higher guest counts and z-scores. Peaks occur sporadically across months, often aligning with weekends, holidays, or special events. These spikes suggest external drivers like promotions, public holidays, or seasonality. Recommendation: Identify patterns behind these high-guest dates (e.g., campaigns, local events) and replicate or plan around them. Additionally, resource planning (staffing, inventory) should be optimized in advance for anticipated peak periods.

- Which month had the highest average number of guests per booking?

```
with Guest as(
select MONTH(ArrivalDate) as MonthNumber,
SUM(Guests) as TotalGuest
from HotelBookings_replica
group by MONTH(ArrivalDate)),
```

```
Booking as (
select MONTH(ArrivalDate) as MonthNumber,
count(BookingID) as Totalbookings
from HotelBookings_replica
group by MONTH(ArrivalDate)
)
```

```
select top(2) g.MonthNumber, cast((g.TotalGuest/b.Totalbookings) as
decimal(10,2)) as AvgGuests
from Guest g
join Booking b on g.MonthNumber=b.MonthNumber
order by AvgGuests desc;
```

MonthNumber	AvgGuests
7	2.00
8	2.00

In July and August (months 7 and 8), the average number of guests per booking was the highest at 2.00, indicating peak travel or holiday season with more group or family bookings. This suggests strong seasonal demand, and the hotel could optimize by offering group packages, early-bird promotions, or higher pricing tiers during these months. It's also a good time to ensure adequate staffing and inventory to handle the increased guest load efficiently.



Forecasting (Trend Detection, Patterns)

1. Calculate month-over-month growth in net revenue.

```
select YearValue,MonthNumber,TotalRev,LagRev,
cast(((TotalRev-LagRev)/LagRev) as decimal(10,2)) as MonthOverMonth
from(
select year(ArrivalDate) as YearValue, MONTH(ArrivalDate) as MonthNumber,
SUM(Revenue) as TotalRev,
LAG(SUM(Revenue)) over (partition by year(ArrivalDate) order by
MONTH(ArrivalDate)) as LagRev
```

from HotelBookings_replica
group by year(ArrivalDate),MONTH(ArrivalDate)) as LagRevByMonth;

YearValue	MonthNumber	TotalRev	LagRev	MonthOverMonth
2015	7	794122.79	NULL	NULL
2015	8	1222882.71	794122.79	0.54
2015	9	1402773.90	1222882.71	0.15
2015	10	1007522.28	1402773.90	-0.28
2015	11	363041.49	1007522.28	-0.64
2015	12	528147.07	363041.49	0.45
2016	1	291521.38	NULL	NULL
2016	2	579198.72	291521.38	0.99
2016	3	919801.53	579198.72	0.59
2016	4	1126280.09	919801.53	0.22
2016	5	1231884.13	1126280.09	0.09
2016	6	1439152.88	1231884.13	0.17

From January to August, revenue steadily increased, with February and July showing the highest growth. However, starting in September, revenue began to decline significantly—dropping almost by half in November. This trend suggests that the summer months (June–August) perform best, so offering promotions during this peak period could maximize revenue. On the other hand, since the year-end months show a drop, strategies like seasonal discounts, festive campaigns, or re-engaging past customers should be used to maintain steady income.

2. What is the 3-month moving average of ADR ?

```
SELECT
YEAR(ArrivalDate) as YearValue,
MONTH(ArrivalDate) as MonthNumber,
AVG(AvgDailyRate) as AvgADR,
ROW_NUMBER() over(
partition by YEAR(ArrivalDate) order by
YEAR(ArrivalDate),MONTH(ArrivalDate),AVG(AvgDailyRate)) as RowNum,
CASE
when ROW_NUMBER() over(
```

```

        partition by YEAR(ArrivalDate) order by
YEAR(ArrivalDate),MONTH(ArrivalDate),AVG(AvgDailyRate))>=3
    then
        AVG(AVG(AvgDailyRate)) over (
        PARTITION BY YEAR(ArrivalDate)
        ORDER BY MONTH(ArrivalDate)
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
        )
    else null
end as 'MovingAvg'
from HotelBookings_replica
group by YEAR(ArrivalDate),MONTH(ArrivalDate)
;

```

YearValue	MonthNumber	AvgADR	RowNum	MovingAvg
2015	7	97.834315	1	NULL
2015	8	105.922887	2	NULL
2015	9	94.818662	3	99.525288
2015	10	78.895426	4	93.212325
2015	11	60.580252	5	78.098113
2015	12	74.079243	6	71.184973
2016	1	64.767655	1	NULL
2016	2	70.102269	2	NULL
2016	3	79.069326	3	71.313083
2016	4	88.918920	4	79.363505
2016	5	96.399286	5	88.129177

The data shows a monthly trend of average guests along with a moving average. From the analysis, we observe that the average number of guests has gradually increased since mid-2015, with a notable rise during the summer months (June to August) in both 2016 and 2017. The moving average clearly indicates an upward trend over time. As a recommendation, the hotel management should capitalize on this seasonal growth by launching targeted promotions, upgrading services, and offering special packages during peak months (especially July and August) to maximize revenue and guest satisfaction.

- Predict next month's revenue using linear trend (past 15 months as input).

WITH BaseTable as(

```

select year(ArrivalDate) as YearValue,
       MONTH(ArrivalDate) as MonthNumber,
       ROW_NUMBER() over(order by year(ArrivalDate) ,MONTH(ArrivalDate))
as MonthIndex,
       AVG(Revenue) as AvgRevnuue,
       (ROW_NUMBER() over(order by year(ArrivalDate)
,MONTH(ArrivalDate))*AVG(Revenue)) as XY,
       POWER(ROW_NUMBER() OVER (ORDER BY YEAR(ArrivalDate),
MONTH(ArrivalDate)), 2) AS X_MultiplyBy_X
from HotelBookings_replica
group by year(ArrivalDate) ,MONTH(ArrivalDate)
),

TotalXY as(
select SUM(XY) as SumXY, SUM(X_MultiplyBy_X) as
SumX_MultiplyBy_X,SUM(MonthIndex) as SumX,
       (SUM(MonthIndex) * SUM(MonthIndex)) as SumX_MultiplyBy_SumX,
SUM(AvgRevnuue) AS SumY
from BaseTable),

calculationTable as (
select b.YearValue,b.MonthNumber,
b.MonthIndex,b.AvgRevnuue,b.XY,b.X_MultiplyBy_X, t.SumXY,
       t.SumX_MultiplyBy_X, t.SumX,t.SumX_MultiplyBy_SumX, t.SumY
from BaseTable b
cross join TotalXY t),

Find_ab as(
select
(((15*c.SumXY)-(c.SumX*c.SumY))/((15*c.SumX_MultiplyBy_X)-(c.SumX_Multiply
By_SumX))) as a,

((c.SumY-(((15*c.SumXY)-(c.SumX*c.SumY))/((15*c.SumX_MultiplyBy_X)-(c.Sum
X_MultiplyBy_SumX))*c.SumX))/15) as b
from calculationTable c)

select distinct (a*16+b) as ExpectedAvgRevForSept2017
from Find_ab;

```

ExpectedAvgRevForSept2017
203.909287

4. Which months show consistent peaks in bookings over the last 3 years (seasonality)?

WITH MonthlyData AS (

```

SELECT
    MONTH(ArrivalDate) AS MonthNumber,
    COUNT(CASE WHEN YEAR(ArrivalDate) = 2015 THEN BookingID END) AS
Bookings_2015,
    COUNT(CASE WHEN YEAR(ArrivalDate) = 2016 THEN BookingID END) AS
Bookings_2016,
    COUNT(CASE WHEN YEAR(ArrivalDate) = 2017 THEN BookingID END) AS
Bookings_2017
FROM HotelBookings_replica
GROUP BY MONTH(ArrivalDate)
),
FULLTABLE AS (
SELECT *,
    (SELECT COUNT(BookingID)/36.0
     FROM HotelBookings_replica) as AvgBookingFor3years
FROM MonthlyData
)

SELECT
    F.MonthNumber,
    CASE WHEN F.Bookings_2015=0 THEN 0 ELSE
CAST(((F.Bookings_2015-F.AvgBookingFor3years)/F.AvgBookingFor3years)*100
AS DECIMAL(10,2)) END AS Seasonality_peak_2015,
    CASE WHEN F.Bookings_2016=0 THEN 0 ELSE
CAST(((F.Bookings_2016-F.AvgBookingFor3years)/F.AvgBookingFor3years)*100
AS DECIMAL(10,2)) END AS Seasonality_peak_2016,
    CASE WHEN F.Bookings_2017=0 THEN 0 ELSE
CAST(((F.Bookings_2017-F.AvgBookingFor3years)/F.AvgBookingFor3years)*100
AS DECIMAL(10,2)) END AS Seasonality_peak_2017
FROM FULLTABLE F
ORDER BY F.MonthNumber;

```

MonthNumber	Seasonality_peak_2015	Seasonality_peak_2016	Seasonality_peak_2017
1	0.00	-32.22	10.99
2	0.00	17.33	25.95
3	0.00	45.46	49.86
4	0.00	63.67	70.70
5	0.00	65.18	90.36
6	0.00	59.57	70.28
7	-16.29	37.86	60.20
8	17.27	52.67	48.50
9	54.20	62.65	0.00
10	49.47	87.04	0.00
11	-29.44	34.30	0.00
12	-11.95	16.39	0.00

The seasonality analysis reveals that bookings in 2015 remained significantly

below average in all months, indicating underperformance or possible data gaps, while 2016 and 2017 show strong seasonal peaks from March to October, especially May to August, suggesting peak tourist demand during these months. Months like September to November in 2017 show a sharp decline, possibly due to external factors or seasonal off-peak trends. Based on this, the hotel should focus marketing and promotional efforts between April and August, scale up staffing and inventory during these peak months, and offer discounts or packages in off-peak months like January, November, and December to balance occupancy and revenue.

5. Compare revenue forecast vs. actual (if you have a StatusUpdate or plan vs actual field).

-- Compare revenue EXPECTED vs. actual (if you have a StatusUpdate or plan vs actual field).

```
SELECT
  MONTH(ArrivalDate) AS MonthNumber,
  (ABS(SUM(RevenueLoss))+SUM(Revenue)) AS ExpectedRevenue,
  SUM(Revenue) AS ActualRevenue,
  ABS(SUM(RevenueLoss)) AS LostRevenue,
  Cast((SUM(Revenue)/(ABS(SUM(RevenueLoss))+SUM(Revenue)))*100 AS
  DECIMAL(10,2)) AS Revenue_Percentage
FROM HotelBookings_replica
GROUP BY MONTH(ArrivalDate)
ORDER BY MONTH(ArrivalDate);
```

MonthNumber	ExpectedRevenue	ActualRevenue	LostRevenue	Revenue_Percentage
1	1273077.43	860798.84	412278.59	67.62
2	1818972.70	1292835.48	526137.22	71.08
3	2639221.09	1913763.65	725457.44	72.51
4	3710466.14	2567842.79	1142623.35	69.21
5	4108125.90	2883721.64	1224404.26	70.20
6	4499387.28	3195043.00	1304344.28	71.01
7	6587713.64	4251957.07	2335756.57	64.54
8	7944717.43	5052890.46	2891826.97	63.60
9	3731917.18	2883533.04	848384.14	77.27
10	3009549.50	2255124.12	754425.38	74.93
11	1600156.54	1213573.16	386583.38	75.84
12	1800192.70	1229514.19	570678.51	68.30

The analysis shows that August and July had the highest potential revenue loss

(approx. 2.9M and 2.3M), resulting in the lowest revenue realization rates of 63.60% and 64.54%, respectively. In contrast, September and November performed better with realization rates above 75%. The recommendation is to implement stricter booking verification, clear cancellation policies, and customer confirmation incentives during high-loss months to reduce the gap between expected and actual revenue.

6. Estimate daily booking trend over time using 7-day rolling averages.

```
SELECT
  YEAR(ArrivalDate) AS ArrivalYear,
  ArrivalDate,
  COUNT(BookingID) AS Daily_Booking,
  ROW_NUMBER() OVER(ORDER BY ArrivalDate) AS RowValue,
  CASE
    WHEN (ROW_NUMBER() OVER(ORDER BY ArrivalDate))>=7 THEN
      AVG(COUNT(BookingID)) OVER(ORDER BY ArrivalDate ROWS BETWEEN 6
      PRECEDING AND CURRENT ROW)
    ELSE NULL END AS '7-day rolling averages'
FROM HotelBookings_replica
GROUP BY YEAR(ArrivalDate),ArrivalDate
ORDER BY YEAR(ArrivalDate),ArrivalDate;
```

ArrivalYear	ArrivalDate	Daily_Booking	RowValue	7-day rolling averages
2015	2015-07-01	122	1	NULL
2015	2015-07-02	93	2	NULL
2015	2015-07-03	56	3	NULL
2015	2015-07-04	88	4	NULL
2015	2015-07-05	53	5	NULL
2015	2015-07-06	75	6	NULL
2015	2015-07-07	54	7	77
2015	2015-07-08	69	8	69

7. Find anomalies in ADR (e.g., days with >2 SD above/below monthly mean).

```
DECLARE @SDForADR DECIMAL(10,2)
DECLARE @MEANForADR DECIMAL(10,2)
```

```
SELECT @SDForADR=STDEV(AvgDailyRate) FROM HotelBookings_replica;
SELECT @MEANForADR=AVG(AvgDailyRate) FROM HotelBookings_replica;
```

```
SELECT AvgDailyRate,
CASE
    WHEN AvgDailyRate>(@MEANForADR+2*@SDForADR) OR
    AvgDailyRate<(@MEANForADR-2*@SDForADR) THEN 'OUTLIER'
    ELSE 'NORMAL'
END AS 'ANOMALIES'
FROM HotelBookings_replica;
```

AvgDailyRate	ANOMALIES
0.00	OUTLIER
0.00	OUTLIER
75.00	NORMAL
75.00	NORMAL
98.00	NORMAL
98.00	NORMAL
107.00	NORMAL
103.00	NORMAL

8. Which customer type shows the most predictable booking pattern (lowest ADR standard deviation)?

```
SELECT
    CustomerType,
    AVG(AvgDailyRate) AS ADR,
    CAST(STDEV(AvgDailyRate) AS DECIMAL(10,2)) AS STD_ADR,
    CAST(STDEV(AvgDailyRate) / NULLIF(AVG(AvgDailyRate), 0) AS
    DECIMAL(10,2)) AS CV_ADR
FROM HotelBookings_replica
GROUP BY CustomerType;
```

CustomerType	ADR	STD_ADR	CV_ADR
Group	83.488578	50.26	0.60
Contract	87.549636	33.69	0.38
Transient	107.013620	53.34	0.50
Transient-Party	86.084253	36.89	0.43

Among all customer types, Contract customers show the most predictable

booking pattern, with the lowest standard deviation (33.69) and lowest coefficient of variation (0.38) in ADR. This indicates their booking rates are stable and consistent. Recommendation: Focus on retaining and expanding Contract customers, as their predictable revenue can help improve forecasting and reduce pricing volatility.

Competitive Analysis (Benchmarking & Comparison)

1. Compare ADR and revenue between hotels if multiple hotels are present.

```
SELECT
    Hotel,
    AVG(AvgDailyRate) AVGADR,
    ROUND(SUM(Revenue)/1000000.0,2) TotalREV
FROM HotelBookings_replica
GROUP BY Hotel;
```

Hotel	AVGADR	TotalREV
Resort	94.952929	12.040000
City	105.304465	17.560000

The City hotel has a higher average daily rate (ADR) than the Resort hotel (105.30 vs 94.95) and also generates more total revenue (17.56M vs 12.04M). This suggests that despite charging higher prices, the City hotel maintains strong customer demand. In contrast, the Resort hotel, with a lower ADR, brings in less revenue, indicating potential underperformance. Recommendation: The Resort hotel should reassess its pricing and marketing strategy—consider introducing off-peak offers, value-added services, or loyalty programs to improve ADR and overall revenue.

2. Which distribution channel yields the highest revenue per guest?

```
SELECT
    DistributionChannel,
    SUM(Revenue)/COUNT(*) RevPerGuests,
    MAX(SUM(Revenue)*1.0/COUNT(*)) OVER() HighestREV
```

**FROM HotelBookings_replica
GROUP BY DistributionChannel;**

DistributionChannel	RevPerGuests	HighestREV
Online Travel Agent	232.950690	302.684874
Corporate	137.560995	302.684874
Offline Travel Agent	291.509135	302.684874
Direct	302.684874	302.684874
Undefined	112.700000	302.684874

The Direct channel brings the most money per guest, earning more than other channels like Offline and Online Travel Agents. Corporate and Undefined channels make much less. This means it's best to focus on getting more bookings directly through your own website or app, and try to improve or rethink how you work with the less profitable channels to earn more overall.

3. Compare cancellation rates across customer types and countries.

**DECLARE @TotalCancelled decimal(10,2)
SELECT @TotalCancelled = SUM(CAST(Cancelled AS decimal)) FROM
HotelBookings_replica;

SELECT
CustomerType,
ISNULL(Country,'UNKNOWN') COUNTRY,
SUM(CAST(Cancelled AS DECIMAL(10,2))) TotalCancellation,
SUM(CAST(Cancelled AS DECIMAL(10,2)))*100/@TotalCancelled
TotalCancellationPercentages
FROM HotelBookings_replica
GROUP BY CustomerType, Country
ORDER BY CustomerType;**

CustomerType	COUNTRY	TotalCancellation	TotalCancellationPercentages
Contract	Romania	0.00	0.000000
Contract	Israel	0.00	0.000000
Contract	Greece	0.00	0.000000
Contract	Croatia	0.00	0.000000
Contract	Luxembourg	0.00	0.000000
Contract	Spain	0.00	0.000000
Contract	Malaysia	0.00	0.000000

4. How does the average lead time of bookings differ between local and international customers(Lets assume UNITED KINGDOM is local)?

```
SELECT
    AVG(CASE
        WHEN Country='United Kingdom' THEN LeadTime ELSE NULL
    END) LocalAVGLeadTime,
    AVG(CASE
        WHEN Country<>'United Kingdom' THEN LeadTime ELSE NULL
    END) InternationalAVGLeadTime
FROM HotelBookings_replica
;
```

LocalAVGLeadTime	InternationalAVGLeadTime
127	101

Local customers (United Kingdom) have a higher average lead time of 127 days, while international customers book with an average lead time of 101 days. This suggests that local guests plan their stays much earlier than international ones. The hotel should consider offering early-bird discounts or loyalty perks to local customers to encourage early planning, while providing flexible and attractive last-minute deals for international guests who tend to book closer to arrival.

5. Determine market share by country (number of bookings).

```
SELECT
    ISNULL(Country,'UNKNOWN') COUNTRY,
    COUNT(BookingID)*100.0/(SELECT COUNT(*) FROM
    HotelBookings_replica WHERE Cancelled<>1) BookingPercentages
FROM HotelBookings_replica
WHERE Cancelled<>1
GROUP BY Country
ORDER BY COUNT(BookingID)*100.0/(SELECT COUNT(*) FROM
    HotelBookings_replica WHERE Cancelled<>1) DESC;
```

COUNTRY	BookingPercentages
Portugal	28.032621131894
United Kingdom	12.872841444270
France	11.283026900460
Spain	8.502514434717
Germany	8.074129260569
Ireland	3.383178564776

Portugal holds the highest market share with 28% of total non-cancelled bookings, indicating a strong local customer base. The majority of bookings come from a few Western European countries like the UK, France, Spain, and Germany. This shows geographic concentration, which may limit growth. To reduce dependency and expand reach, it's recommended to diversify by targeting underrepresented international markets through regional marketing strategies and partnerships.

6. Compare top 5 customer segments by net profit per booking.

```
SELECT
    CustomerType,
    SUM(Revenue) TotalREV,
    ABS(SUM(RevenueLoss)) TotalREVLoss,
    SUM(Revenue)-ABS(SUM(RevenueLoss)) NetProfit,
    CAST(SUM(Revenue) - ABS(SUM(RevenueLoss)) AS FLOAT) /
    NULLIF(COUNT(*), 0) AS ProfitPerBooking
FROM HotelBookings_replica
GROUP BY CustomerType
ORDER BY SUM(Revenue)-ABS(SUM(RevenueLoss)) DESC;
```

CustomerType	TotalREV	TotalREVLoss	NetProfit	ProfitPerBooking
Transient	22696813.88	11502290.03	11194523.85	124.920757590975
Transient-Party	5225239.33	1319396.25	3905843.08	155.462628562331
Contract	1555189.13	283888.62	1271300.51	311.899045632974
Group	123355.10	17325.19	106029.91	183.760675909879

The Contract segment generates the highest net profit per booking (≈312) despite having lower total revenue, indicating it is the most cost-effective and profitable group. Transient-Party and Group customers also show strong profit per booking (≈155 and ≈184), while Transient has the lowest profit per booking (≈125) despite contributing the most to total revenue. Recommendation: Focus on retaining and expanding Contract bookings, while optimizing costs or pricing

for Transient customers to boost profitability.



Bonus: Cross-Domain Analytical Questions

1. What is the correlation between lead time and cancellation rate?

```
SELECT
CASE
  WHEN LeadTime BETWEEN 0 AND 73 THEN 1
  WHEN LeadTime BETWEEN 74 AND 147 THEN 2
  WHEN LeadTime BETWEEN 148 AND 221 THEN 3
  WHEN LeadTime BETWEEN 222 AND 295 THEN 4
  WHEN LeadTime BETWEEN 296 AND 369 THEN 5
  WHEN LeadTime BETWEEN 370 AND 443 THEN 6
  WHEN LeadTime BETWEEN 444 AND 517 THEN 7
  WHEN LeadTime BETWEEN 518 AND 591 THEN 8
  WHEN LeadTime BETWEEN 592 AND 665 THEN 9
  WHEN LeadTime BETWEEN 666 AND 737 THEN 10
  ELSE 'Unknown'
END AS LeadTimeBinList,
CASE
  WHEN LeadTime BETWEEN 0 AND 73 THEN '0-73'
  WHEN LeadTime BETWEEN 74 AND 147 THEN '74-147'
  WHEN LeadTime BETWEEN 148 AND 221 THEN '148-221'
  WHEN LeadTime BETWEEN 222 AND 295 THEN '222-295'
  WHEN LeadTime BETWEEN 296 AND 369 THEN '296-369'
  WHEN LeadTime BETWEEN 370 AND 443 THEN '370-443'
  WHEN LeadTime BETWEEN 444 AND 517 THEN '444-517'
  WHEN LeadTime BETWEEN 518 AND 591 THEN '518-591'
  WHEN LeadTime BETWEEN 592 AND 665 THEN '592-665'
  WHEN LeadTime BETWEEN 666 AND 737 THEN '666-737'
  ELSE 'Unknown'
END AS LeadTimeBin,
COUNT(*) AS TotalBookings,
SUM(CAST(Cancelled AS INT)) AS Cancellations,
ROUND(SUM(CAST(Cancelled AS FLOAT))*100.0 / COUNT(*), 2) AS
CancellationRate
FROM HotelBookings_replica
GROUP BY
CASE
  WHEN LeadTime BETWEEN 0 AND 73 THEN '0-73'
  WHEN LeadTime BETWEEN 74 AND 147 THEN '74-147'
```



```

WHEN LeadTime BETWEEN 148 AND 221 THEN '148-221'
WHEN LeadTime BETWEEN 222 AND 295 THEN '222-295'
WHEN LeadTime BETWEEN 296 AND 369 THEN '296-369'
WHEN LeadTime BETWEEN 370 AND 443 THEN '370-443'
WHEN LeadTime BETWEEN 444 AND 517 THEN '444-517'
WHEN LeadTime BETWEEN 518 AND 591 THEN '518-591'
WHEN LeadTime BETWEEN 592 AND 665 THEN '592-665'
WHEN LeadTime BETWEEN 666 AND 737 THEN '666-737'
ELSE 'Unknown'
END,
CASE
WHEN LeadTime BETWEEN 0 AND 73 THEN 1
WHEN LeadTime BETWEEN 74 AND 147 THEN 2
WHEN LeadTime BETWEEN 148 AND 221 THEN 3
WHEN LeadTime BETWEEN 222 AND 295 THEN 4
WHEN LeadTime BETWEEN 296 AND 369 THEN 5
WHEN LeadTime BETWEEN 370 AND 443 THEN 6
WHEN LeadTime BETWEEN 444 AND 517 THEN 7
WHEN LeadTime BETWEEN 518 AND 591 THEN 8
WHEN LeadTime BETWEEN 592 AND 665 THEN 9
WHEN LeadTime BETWEEN 666 AND 737 THEN 10
ELSE 'Unknown'
END
ORDER BY LeadTimeBinList;

```

LeadTimeBinList	LeadTimeBin	TotalBookings	Cancellations	CancellationRate
1	0-73	61639	15723	25.51
2	74-147	24403	10380	42.54
3	148-221	16345	7539	46.12
4	222-295	8766	4961	56.59
5	296-369	5157	3545	68.74
6	370-443	1862	1150	61.76
7	444-517	824	580	70.39
8	518-591	230	184	80
9	592-665	162	162	100
10	666-737	2	0	0

As lead time increases, the cancellation rate rises sharply—from 25.51% in the 0–73 day bin to 100% in the 592–665 day bin, indicating a strong positive correlation between longer lead times and higher cancellation risk. Most bookings are concentrated in the lower bins, but those with longer lead times (over 200 days) show significantly higher cancellation rates, often exceeding 70%. Recommendation: Implement stricter cancellation policies, deposits, or incentives for long lead-time bookings to reduce cancellations and revenue loss. Also, consider overbooking strategies or focused follow-ups for these high-risk segments.

2. What percentage of cancelled bookings came from customers with >150 & <150 day lead time?

```
SELECT
    SUM(CASE
        WHEN LeadTime>150 THEN cast(Cancelled as decimal) ELSE 0
    END) AS '>150 & <150',
    SUM(cast(Cancelled as decimal)) TotalCancellation,
    SUM(CASE
        WHEN LeadTime>150 THEN cast(Cancelled as decimal) ELSE 0
    END)*100.0/SUM(cast(Cancelled as decimal)) Cancellationpercentages
FROM HotelBookings
UNION ALL
SELECT
    SUM(CASE
        WHEN LeadTime<150 THEN cast(Cancelled as decimal) ELSE 0
    END) AS '>150 & <150',
    SUM(cast(Cancelled as decimal)) TotalCancellation,
    SUM(CASE
        WHEN LeadTime<150 THEN cast(Cancelled as decimal) ELSE 0
    END)*100.0/SUM(cast(Cancelled as decimal)) Cancellationpercentages
FROM HotelBookings;
```

>150 & <150	TotalCancellation	Cancellationpercentages
17678	44224	39.973769
26380	44224	59.650868

Although customers with lead times less than 150 days account for the largest number of cancellations (about 60%), those with lead times greater than 150 days still represent a significant portion (around 40%). This suggests that both short-term and long-term bookings have cancellation risks. Recommendation: Focus on improving booking commitment strategies for short lead times, such as stricter cancellation fees or reminders, while also offering flexible options for long lead-time customers to balance revenue and customer satisfaction.

3. Is there a statistical relationship between number of guests and ADR (use grouped SQL averages)?
4. create a pivot table for CustomerType for various Aggregate calculations?

```
SELECT
    'Booking Count' AS PivotColumn,
```

```

*
FROM (
    SELECT
        BookingID,
        CustomerType
    FROM HotelBookings_replica
) AS BookingCount
PIVOT (
    COUNT(BookingID)
    FOR CustomerType IN ([Transient], [Group], [Contract], [Transient-Party])
) AS BookingCountPivot

UNION ALL

SELECT
    'Avg ADR' AS PivotColumn,
    *
FROM (
    SELECT
        AvgDailyRate,
        CustomerType
    FROM HotelBookings_replica
) AS AvgADR
PIVOT (
    AVG(AvgDailyRate)
    FOR CustomerType IN ([Transient], [Group], [Contract], [Transient-Party])
) AS AvgADRPivot

UNION ALL

SELECT
    'Total Revenue' AS PivotColumn,
    *
FROM (
    SELECT
        Revenue,
        CustomerType
    FROM HotelBookings_replica
) AS AvgADR
PIVOT (
    SUM(Revenue)
    FOR CustomerType IN ([Transient], [Group], [Contract], [Transient-Party])
) AS TotalRevenuePivot;

```

PivotColumn	Transient	Group	Contract	Transient-Party
Booking Count	89613.00	577.00	4076.00	25124.00
Avg ADR	107.01	83.49	87.55	86.08
Total Revenue	22696813.88	123355.10	1555189.13	5225239.33

5. Calculate z-scores for ADR to detect rate outliers.

```
DECLARE @Mean FLOAT
SELECT @Mean =AVG(AvgDailyRate) FROM HotelBookings_replica;
DECLARE @STDDEV FLOAT
SELECT @STDDEV = STDEV(AvgDailyRate) FROM HotelBookings_replica;

WITH ZSCORE AS (
SELECT
    AvgDailyRate ,
    ((AvgDailyRate-@Mean)/@STDDEV) AS [Z-SCORE]

FROM HotelBookings_replica
)

SELECT
    *,
    CASE
        WHEN [Z-SCORE]>2 THEN 1
        WHEN [Z-SCORE]<-2 THEN 1
        ELSE 0
    END AS FLAG
FROM ZSCORE;
```

AvgDailyRate	Z-SCORE	FLAG
0.00	-2.0150297526686	1
0.00	-2.0150297526686	1
75.00	-0.530933044648024	0
75.00	-0.530933044648024	0